

Step-by-Step Protocol for Generating Wolffia Transcriptomic Data

Pilot-to-discovery workflow for generating a first *Wolffia australiana* single-cell or single-nucleus transcriptomic dataset.

This document is written as a clean, stepwise experimental protocol. It is designed for the first Wolffia data-generation effort and emphasizes route selection between intact cells and nuclei before scaling to a larger sequencing run.

Purpose

The purpose of this protocol is to generate a first Wolffia dataset that can recover broad transcriptional programs reproducibly, distinguish biological structure from preparation artifacts, and support the first Wolffia-native atlas analysis.

The key operational question is whether Wolffia yields cleaner transcriptomes through intact-cell preparation or nuclei preparation.

Experimental Scope

- Wolffia culture standardization
- Pre-harvest culture quality control
- Matched pilot comparison of intact-cell and nuclei routes
- Route-selection criteria
- Execution of the first full discovery run
- Immediate post-sequencing quality control

Recommended Baseline Culture Framework

If a successful local Wolffia maintenance condition already exists, use that condition without unnecessary re-optimization during the pilot phase. If no standard exists, use the following pilot baseline for consistency:

- Axenic liquid culture if possible
- One Wolffia line or stock only
- One baseline vegetative growth condition
- 16-hour light / 8-hour dark photoperiod
- 22 to 25 C stable growth temperature
- Low-to-moderate constant light appropriate for healthy duckweed growth
- Harvest at the same time of day for all replicates

Required Materials and Equipment

Category	Key items
Culture and sampling	Healthy Wolffia stock, growth medium, sterile vessels, sterile pipettes and tips, sterile forceps or sampling tools, metadata sheet or electronic log
Pre-harvest QC	Brightfield microscope or stereomicroscope, imaging device, optional balance for wet-weight estimation
Intact-cell pilot	Fine blade or gentle disruption tool, osmotic stabilization buffer, plant cell-wall digestion reagents, mesh filters or strainers, low-speed centrifuge, hemocytometer or automated counter, optional viability dye

Category	Key items
Nuclei pilot	Ice bucket or cold block, nuclei isolation buffer, gentle homogenization setup such as a Dounce homogenizer, mesh filters or strainers, refrigerated centrifuge, nuclei counting setup, optional DNA stain
Library preparation and sequencing	Chosen library-prep kit or core-facility workflow, low-bind tubes, RNase-free plastics, PCR or amplification equipment, library quantification system, sequencing core or sequencing platform access

Required Metadata

- Sample ID
- Biological replicate ID
- Stock or genotype ID
- Medium
- Temperature
- Photoperiod
- Harvest date and time
- Operator
- Preparation route: cell or nucleus
- Preparation condition ID
- Viability or nuclei-integrity observation
- Library batch
- Sequencing batch

Protocol

Step 1: Establish and stabilize Wolffia cultures

Goal: Create standardized starting material before any pilot preparation begins.

Actions:

- 1 Start with one Wolffia line or stock only.
- 2 Grow the material under one defined baseline vegetative condition.
- 3 Maintain the same medium, light schedule, and temperature across all pilot cultures.
- 4 Avoid mixing very old and very young cultures in the same experimental batch.
- 5 Keep a culture log for at least several days before harvest.

QC or decision points:

- Cultures should be visibly healthy and not overcrowded.
- No obvious contamination should be present.
- Growth conditions should be consistent across vessels.

Expected outputs:

- Healthy, standardized Wolffia cultures
- Stable maintenance conditions
- Clear metadata records

Step 2: Perform pre-harvest culture QC

Goal: Confirm that the material is worth processing.

Actions:

- 1 Inspect representative fronds by brightfield microscopy or stereomicroscopy.
- 2 Record visible morphology, frond size, budding status, and contamination.
- 3 Estimate available biomass by frond count, wet weight, or both.
- 4 Exclude cultures that are obviously stressed, contaminated, senescent, or highly heterogeneous.

QC or decision points:

- Proceed only if cultures are visibly healthy, reasonably uniform, and sufficient in biomass for a split pilot.

Expected outputs:

- Microscopy images
- Biomass estimate
- Go or no-go decision for each planned replicate

Step 3: Harvest fresh Wolffia tissue

Goal: Collect material in a way that minimizes artificial damage.

Actions:

- 1 Harvest tissue gently to avoid mechanical damage.
- 2 Use the same harvest method and timing for all matched pilot samples.
- 3 Minimize the time between harvest and the start of preparation.
- 4 For nuclei preparations, place tissue on ice immediately after harvest.
- 5 For intact-cell preparations, move tissue rapidly into the chosen wash or stabilization buffer.

QC or decision points:

- Record harvest-to-processing time for every sample.
- Do not allow one sample to sit substantially longer than another.

Expected outputs:

- Fresh matched material ready for route comparison
- Documented handling times

Step 4: Split the matched pilot into two preparation routes

Goal: Create a fair technical comparison between the cell and nuclei approaches.

Actions:

- 1 From the same starting biological material, divide the sample into Route A and Route B.
- 2 Assign Route A to intact-cell preparation.
- 3 Assign Route B to nuclei preparation.

QC or decision points:

- This split is for technical comparison, not biological comparison.
- The starting material for both routes should be as similar as possible.

Expected outputs:

- Matched pilot material for the two routes

Step 5A: Run the intact-cell preparation pilot

Goal: Test whether intact, viable single cells can be recovered with acceptable quality.

Actions:

- 1 Rinse freshly harvested *Wolffia* briefly to remove residual medium.
- 2 Disrupt the tissue gently using a fine blade or similarly mild mechanical method.
- 3 Transfer tissue into a plant cell-wall digestion solution.
- 4 Incubate under gentle conditions for a short pilot time course.
- 5 Filter the digest through a suitable mesh to remove large debris.
- 6 Wash the recovered cells into an osmotic stabilization buffer.
- 7 Count cells immediately and inspect morphology.
- 8 Measure viability if the workflow supports it.

QC or decision points:

- Compare a small pilot matrix rather than assuming one final recipe on day one.
- Track total recovered cell count, concentration, viability, visible clumping, debris burden, and time from harvest to stabilization.
- Failure criteria include mostly broken material, very low viability, extreme clumping, very low yield, or harsh treatment required just to recover any cells.

Expected outputs:

- Cell yield and viability measurements
- Morphology notes
- Evidence for or against continuing the cell route

Step 5B: Run the nuclei preparation pilot

Goal: Test whether nuclei provide a cleaner and more reproducible Wolffia input.

Actions:

- 1 Keep freshly harvested Wolffia cold from the start of processing.
- 2 Transfer tissue into a cold nuclei isolation buffer.
- 3 Homogenize gently just enough to release nuclei.
- 4 Filter the homogenate to remove large debris.
- 5 Pellet and wash nuclei carefully under cold conditions.
- 6 Count nuclei and inspect nuclear integrity.
- 7 If appropriate, use a standard nuclei stain for counting or gating.

QC or decision points:

- Track nuclei count, clumping, visible nuclear integrity, debris burden, apparent plastid carryover, and apparent ambient contamination.
- Failure criteria include extensive rupture, severe aggregation, extreme contamination, or very poor recovery.

Expected outputs:

- Nuclei yield and integrity measurements
- Evidence for or against continuing the nuclei route

Step 6: Choose the route for the full experiment

Goal: Select the best upstream route using evidence rather than preference.

Actions:

- 1 Review the intact-cell and nuclei pilot outputs side by side.
- 2 Select the intact-cell route only if viable cell recovery is reproducible and debris, clumping, and harshness remain manageable.
- 3 Select the nuclei route if the cell route is harsh, inconsistent, or low-yield and nuclei are clearly cleaner and more reproducible.

QC or decision points:

- Do not force the intact-cell route if nuclei are clearly superior.
- Choose the route most likely to preserve interpretable biology.

Expected outputs:

- One selected preparation route for the full discovery run

Step 7: Run the first full *Wolffia* discovery experiment

Goal: Generate the first real *Wolffia* dataset under a simple, controlled design.

Actions:

- 1 Lock the upstream preparation route chosen from the pilot.
- 2 Run one defined vegetative condition only.
- 3 Collect three biological replicates.
- 4 Use one fixed preparation route and one fixed library-prep workflow.

QC or decision points:

- Do not combine multiple perturbations or rare-state enrichment schemes in the first run.
- The first experiment should remain simple and interpretable.

Expected outputs:

- A balanced first discovery dataset
- Three biological replicates processed under one route

Step 8: Prepare libraries and sequence

Goal: Generate libraries in a balanced and consistent way.

Actions:

- 1 Use the selected input type consistently across all replicates.
- 2 Keep library preparation as uniform as possible across replicates.
- 3 Sequence replicates in a balanced way to avoid obvious batch imbalance.
- 4 Prioritize library quality and replicate consistency over scale inflation.

QC or decision points:

- If the lab uses PIP-seq, upstream preparation quality, low debris, and the correct loading range remain the key *Wolffia*-specific variables.
- If the lab uses another lower-throughput full-length workflow or a higher-throughput nuclei-compatible workflow later, the same principle still applies.

Expected outputs:

- Sequence-ready libraries
- Balanced sequencing submission

Step 9: Perform immediate post-sequencing QC

Goal: Determine whether the experiment produced useful and interpretable data.

Actions:

- 1 Evaluate reads per cell or per nucleus.
- 2 Evaluate genes detected per cell or per nucleus.
- 3 Measure organellar fraction.
- 4 Assess ambient-RNA-like background.
- 5 Check replicate mixing.
- 6 Check stress-like artifact burden.
- 7 Assess the stability of recovered clusters or neighborhoods.

QC or decision points:

- Minimum biological success is recovery of at least some broad expected programs in a reproducible way.
- Examples include proliferative or meristematic signal, photosynthetic or assimilation signal, transport or interface-associated signal, and developmental transition-like signal.

Expected outputs:

- Immediate QC summary
- Evidence that the first *Wolffia* run is biologically usable or needs revision

What Not to Do

- Do not change culture conditions immediately before harvest.
- Do not process stressed or mixed-quality material.
- Do not force an intact-cell route if nuclei are clearly cleaner.
- Do not over-optimize indefinitely without generating a real test dataset.
- Do not over-interpret technical stress as novel *Wolffia* biology.

Recommended Timeline

Week	Main goal
1	Standardize cultures and finalize metadata tracking.
2	Run matched intact-cell and nuclei pilots.
3	Choose the cleaner route and repeat on fresh material if needed.
4	Prepare the first full libraries and submit sequencing.
5+	Run computational QC and first-pass biological interpretation.

References

- Park J. et al. 2021. Genome of the world's smallest flowering plant, *Wolffia australiana*, helps explain its specialized physiology and unique morphology. *Communications Biology*. doi:10.1038/s42003-021-02389-w
- Hoang P.T.N. et al. 2022. The genome of *Wolffia australiana* facilitates discovery of genetic basis for aquatic adaptation in duckweeds. *The Plant Cell*. doi:10.1093/plcell/koac068

- Public Wolffia reference datasets already identified in this project and useful for assay benchmarking:
PRJNA1124135 Wolffia scRNA-seq and PRJNA809022 Wolffia snRNA-seq

Reference use note: the papers above provide the biological context for Wolffia as a reduced but developmentally competent flowering plant. The exact dissociation chemistry for the first pilot should still be treated as a laboratory optimization variable unless a validated Wolffia or closely related duckweed workflow already exists locally.